

Discriminant Analysis

Method Dossier with Field-Specific Examples | Standard Level

1. Method Description

Discriminant analysis classifies objects into known groups. Goal: find a linear (LDA) or quadratic (QDA) function separating groups optimally. $D = b_1x_1 + b_2x_2 + \dots + b_kx_k$.

LDA assumes equal covariance matrices across groups; QDA allows group-specific covariances. The discriminant function maximizes between-group vs. within-group variance. Wilks' Lambda tests significance.

2. Field-Specific Examples

Social Sciences

Predicting study dropouts vs. graduates using GPA, satisfaction, finances. LDA, $n=500$, hit rate 74%.

Economics

Credit default prediction using income, debt, age. LDA, 82% accuracy.

Medicine

Diabetes diagnosis from blood values. QDA, sensitivity 89%, specificity 91%.

Humanities

Authorship attribution via stylometric features. LDA, 97% correct classification.

Sports Sciences

Talent scouting: future pros vs. amateurs using fitness tests. LDA, 81% hit rate.

3. Assumptions and Prerequisites

- Metric predictors (binary/ordinal with 5+ levels acceptable)
- Known group membership (nominal/binary)
- Multivariate normality per group
- LDA: homogeneity of covariance matrices (Box's M test, $p > .05$)
- QDA: no homogeneity needed, larger samples needed
- No perfect multicollinearity
- Minimum $n > \text{number of predictors per group}$

4. Interpretation and Reporting

- Wilks' Lambda + χ^2 test
- Eigenvalue λ + canonical correlation
- Standardized coefficients (variable importance)
- Structure matrix (correlations with discriminant function)
- Confusion matrix + hit rate (original + cross-validated)
- Sensitivity, specificity, predictive values

5. Code Examples

R — LDA with MASS package:

```
library(MASS)
lda_mod <- lda(g ~ x1+x2, data=d)
pred <- predict(lda_mod)
table(d$g, pred$class)
lda_cv <- lda(g ~ x1+x2, data=d, CV=TRUE)
```

Python — LDA with scikit-learn:

```
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis
lda = LinearDiscriminantAnalysis(); lda.fit(X,y)
y_pred = lda.predict(X)
from sklearn.metrics import confusion_matrix
```

SPSS:

```
DISCRIMINANT /GROUPS=g(0 1) /VARIABLES=x1 x2 /CLASSIFY=NONMISSING POOLED.
```

6. Extension

Multinomial / Canonical Discriminant Analysis

For 3+ groups, multiple discriminant functions extracted. Functions 1, 2, ... separate residual variance.

Regularized Discriminant Analysis (RDA)

LDA/QDA hybrid. Useful for $p > n$ situations (e.g., microarray data).

Logistic Regression as Alternative

More flexible: no normality assumption, direct probabilities. LDA better with normal predictors and small samples.

7. Exercises

Exercise 1: Run LDA on iris (Species ~ 4 predictors). How many discriminant functions? Hit rate? Best predictors?

Exercise 2: Bank n=200 (30 defaults). LDA: Wilks' $\Lambda = 0.52$, $p < .001$. Calculate canonical correlation. Interpret.

Exercise 3: When QDA vs. LDA? When logistic regression vs. discriminant analysis? Create a decision matrix.